CarbonLens

"Turning energy data into climate insight."

PROJECT BY

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Problem We’re Solving

- Modern appliances consume large amounts of electricity, but users have **little visibility** into their true energy impact.
- Carbon emissions from everyday household appliances are **difficult to estimate** without data-driven tools.
- There is **no simple way to compare** appliances by energy efficiency or predict their future carbon footprint.
- Consumers and policymakers need **clearer insights** to make informed energy and sustainability decisions.

OUR GOAL

"Analyze appliance energy data and uncover patterns that help predict carbon and energy impact."

Cleaning Up the Data

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Cleaning up the data is pretty straightforward. Here's the formula we applied to prepare our datasets for analysis:

1. **Drop Duplicates**
   Remove repeated entries
2. **Remove Empty Columns**
   Drop completely empty columns
3. **Remove Missing Key Values**
   Drop rows with critical nulls
4. **Convert Numeric Columns**
   Ensure proper data types
5. **Drop Invalid Energy Values**
   Remove rows with 0 or invalid energy
6. **Standardize Brand Names**
   Normalize naming conventions
7. **Convert Yes/No to 1/0**
   Binary encoding for flags
8. **Reset the Index**
   Clean sequential indexing
9. **Convert to Datetime**
   Parse date columns properly
10. **Save to File**
    Export cleaned dataset

This standardized pipeline ensures all datasets are clean, consistent, and ready for exploratory analysis.

Medallion Architecture

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Exploratory Data Analysis - Shri

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What is EDA?

Exploratory Data Analysis (EDA) in machine learning is a process used to **analyze and visualize datasets** to understand their main characteristics, identify patterns, and spot anomalies before applying more advanced analysis techniques.

- Helps ensure that the data is **clean and suitable** for modeling
- This ensures **accurate prediction**

"Let's begin by applying EDA to our cleaned appliance datasets..."

EDA: Refrigerators Dataset

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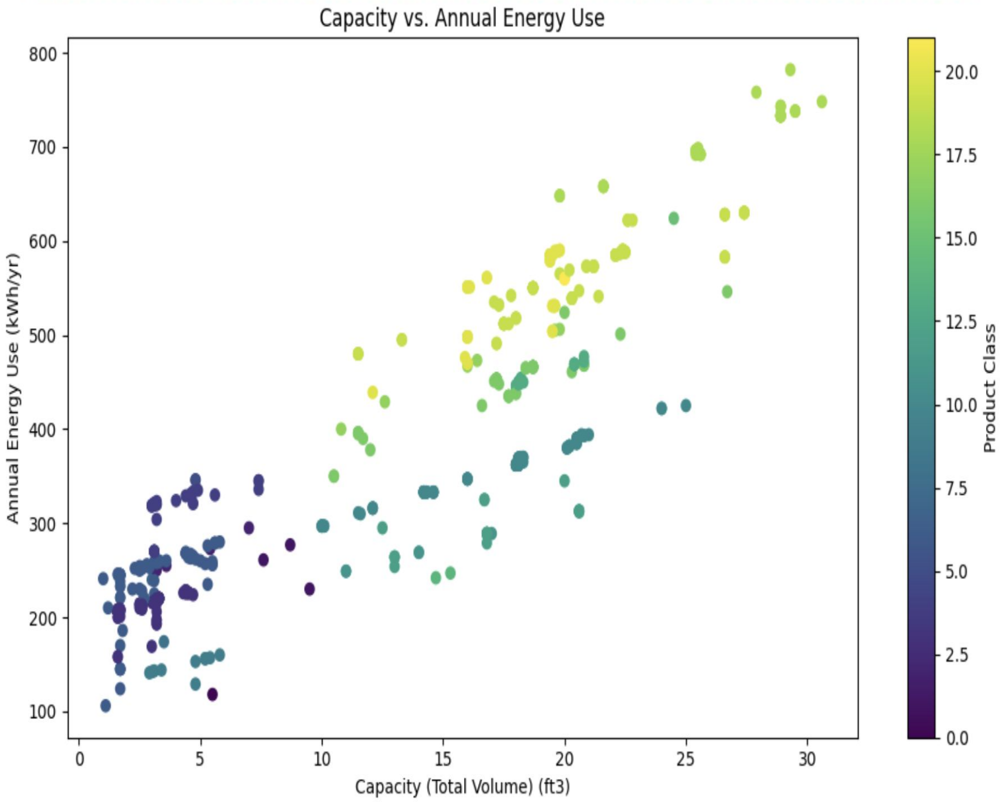
604 rows | **32** features

Down from 4,398 rows before cleaning

KEY FINDINGS

- **Strong positive trend:** Larger refrigerators consume more energy
- **Clear clusters** based on product classes — efficiency varies by design
- **Outliers detected:** Some small fridges use high energy; some large ones are efficient



> **Conclusion: Refrigerator energy consumption is strongly size-dependent.**

EDA: Air Conditioners Dataset

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255 rows | **28** features

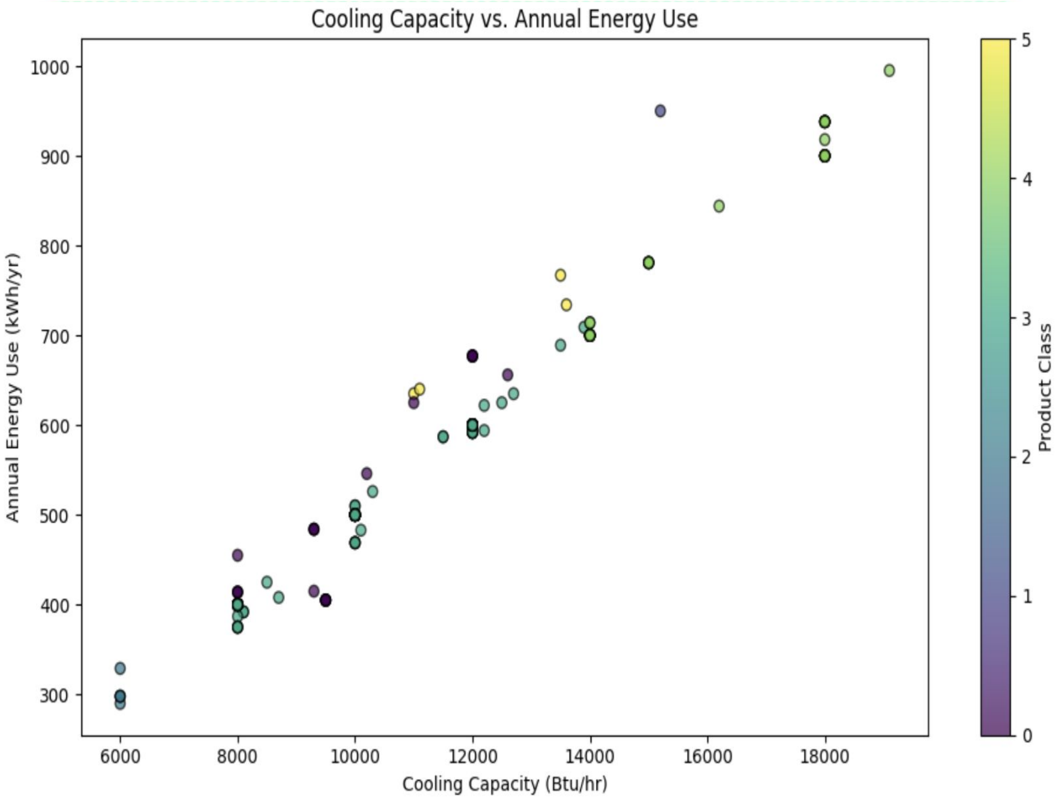
Cleaned from 282 rows (removed missing key values)

Btu/hr (British thermal units per hour) — measures rate of heat transfer in cooling systems

KEY FINDINGS

- **Clear strong positive relationship:** Higher cooling capacity = more energy
- **Distinct clusters** by product class (window, portable, through-the-wall)
- Fewer dramatic outliers compared to refrigerators

Conclusion: Energy consumption scales directly with cooling capacity.



EDA: Televisions Dataset

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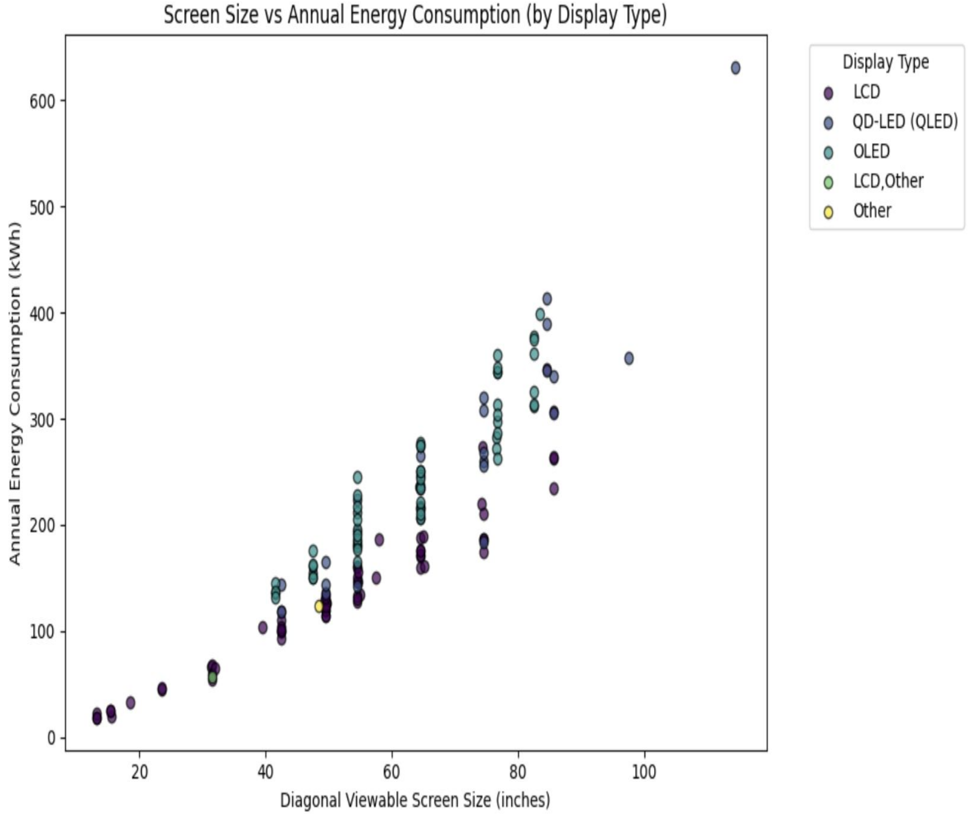
146 rows | **35** columns

All rows retained — key numerical fields were complete

KEY FINDINGS

- **Clear positive trend:** Larger screens require more power
- **Natural clusters** around standard sizes: 40", 55", 65", 75"
- **Display tech matters:** OLED tends to consume more at certain sizes
- Outliers: Some large LCD/QLED TVs draw disproportionately high energy

Conclusion: Screen size is the strongest predictor; display technology adds variation.



EDA: Ceiling Fans Dataset

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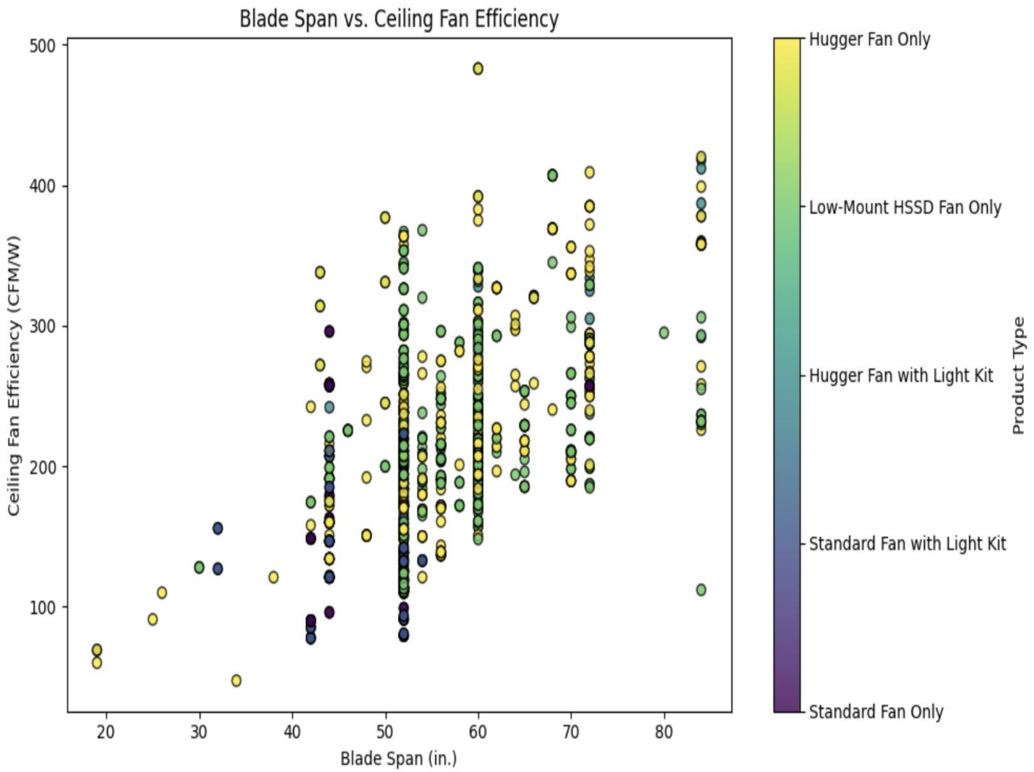
1,144 rows retained

Blade span & ceiling fan efficiency complete and usable

KEY FINDINGS

- **General trend:** Larger fans tend to operate more efficiently
- **Weak relationship:** Not as tight as previous datasets
- **High variation** in mid-range sizes — design choices matter
- Product type influences efficiency patterns significantly



Conclusion: Fan efficiency isn't size-dependent alone — product design plays a major role.

EDA: Ovens Dataset

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Focus Variables:

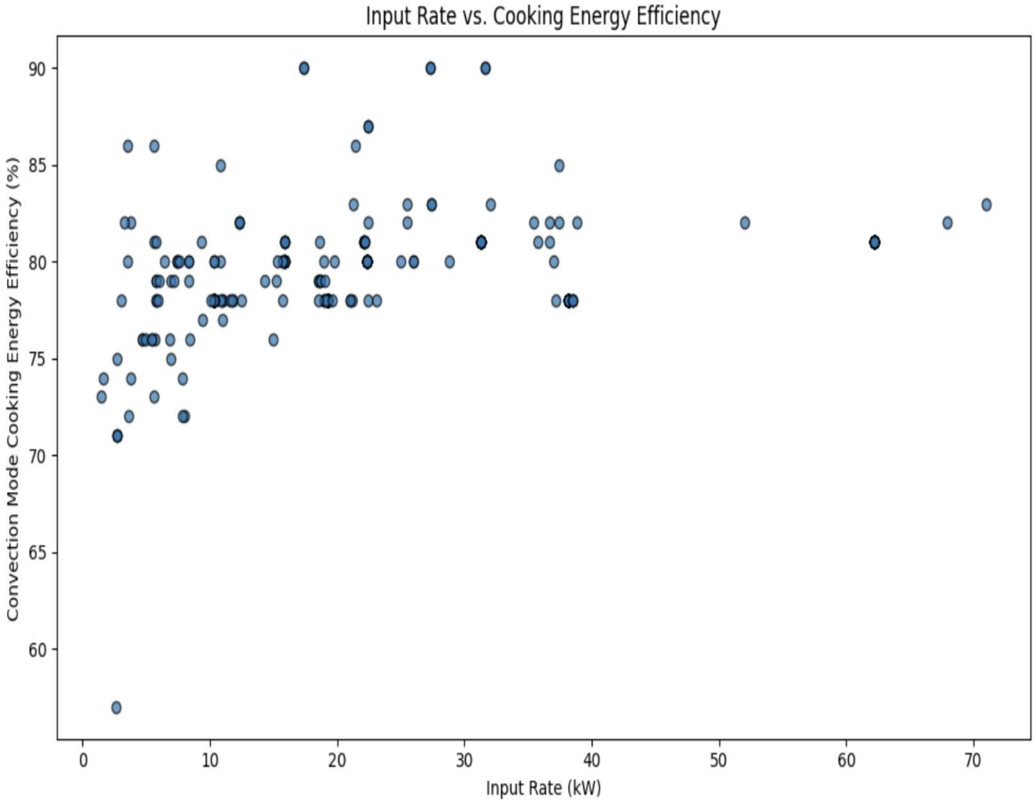
Input Rate (kW) **Convection Efficiency** (%)

Units corrected from Btu/hr, extreme outliers removed

KEY FINDINGS

- **No strong relationship** between power input and cooking efficiency
- Most ovens cluster within **70–90% efficiency** range
- Some outliers: high power but below-average efficiency
- Lower-input ovens can perform exceptionally well (better insulation/heating)

Conclusion: Efficiency is design-driven, not power-driven. Most convection ovens operate at similar efficiency levels.



Television Visualizations

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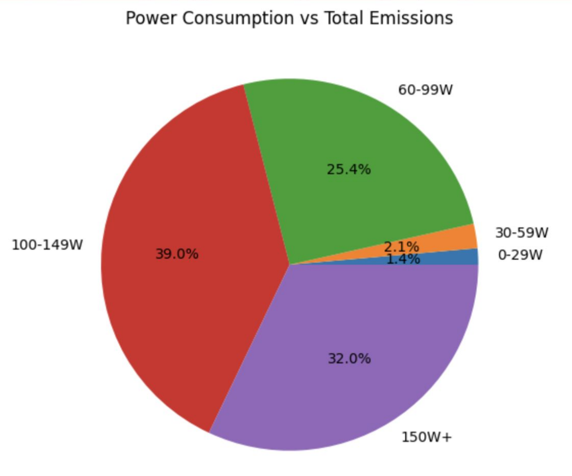
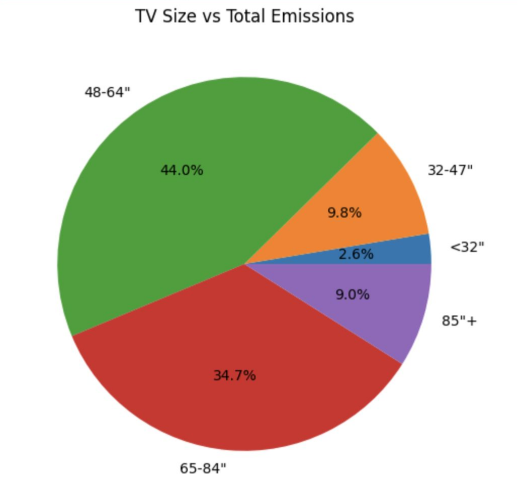
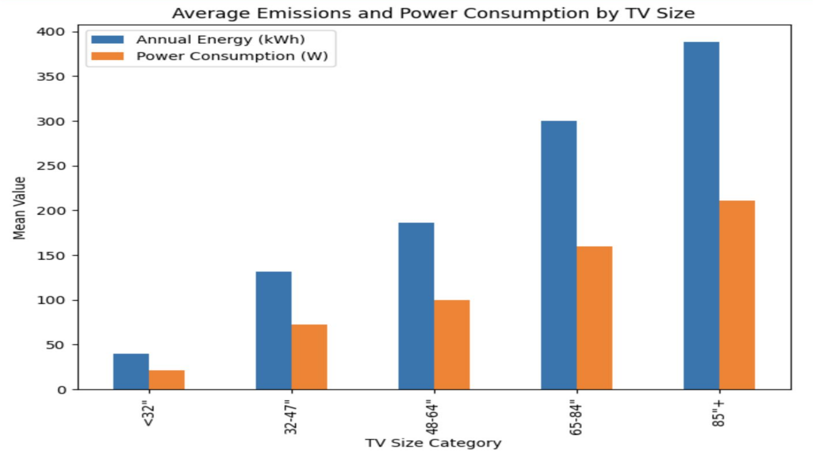
- **Data balance confirmed**: Pie charts showed reasonable distribution across TV size categories, ensuring unbiased analysis
- **Larger TVs = higher emissions**: Bar graphs revealed a clear upward trend in energy usage as screen size increases
- **Significant environmental impact**: Larger TVs consume substantially more power than smaller models
- **Hypothesis supported**: The data confirms that TV size directly correlates with environmental footprint







Ceiling Fans Visualizations

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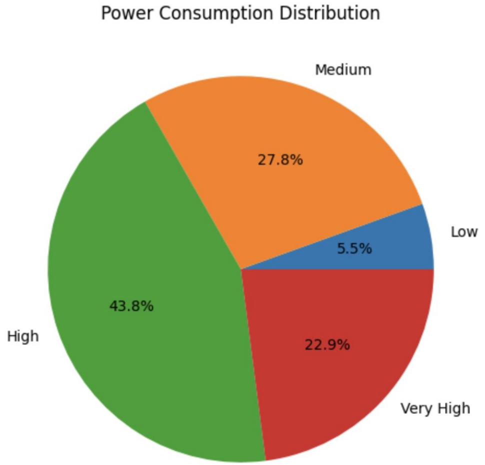
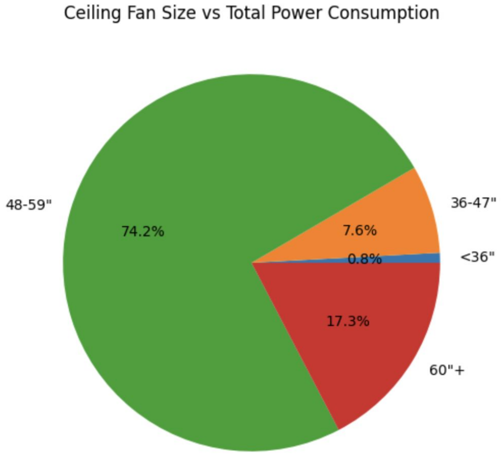
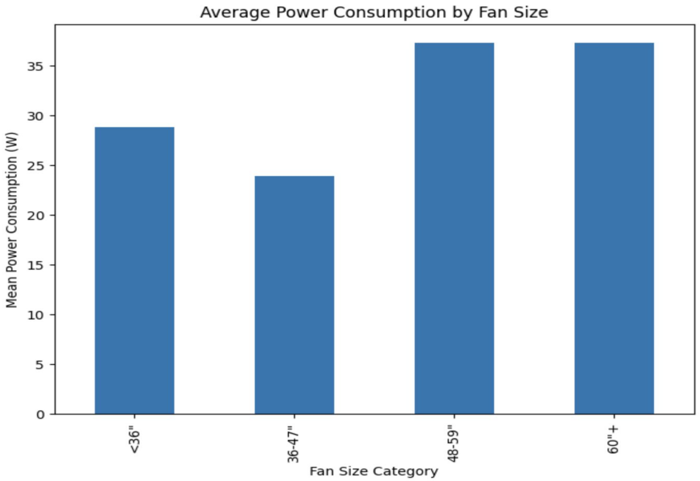
- **Data balance check**: Pie charts revealed some imbalance with 48–59" fans dominating power consumption, but sufficient representation across sizes for meaningful analysis
- **Hypothesis validated**: Bar chart confirmed that average power consumption increases with fan size
- **Largest fans consume most**: 60"+ fans showed the highest mean wattage across all size categories
- **Size matters for efficiency**: Fan size is a critical factor when evaluating energy efficiency and environmental impact

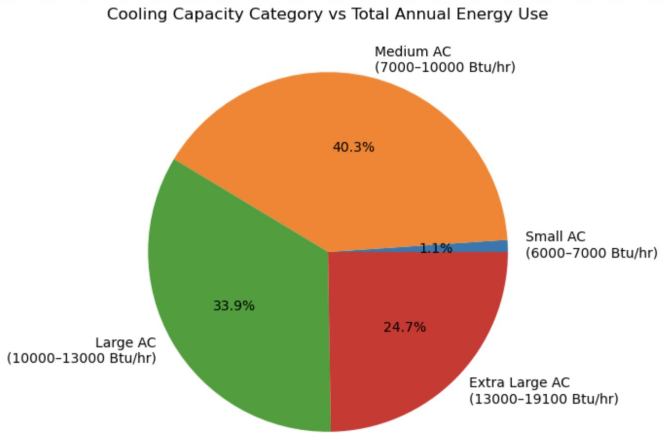


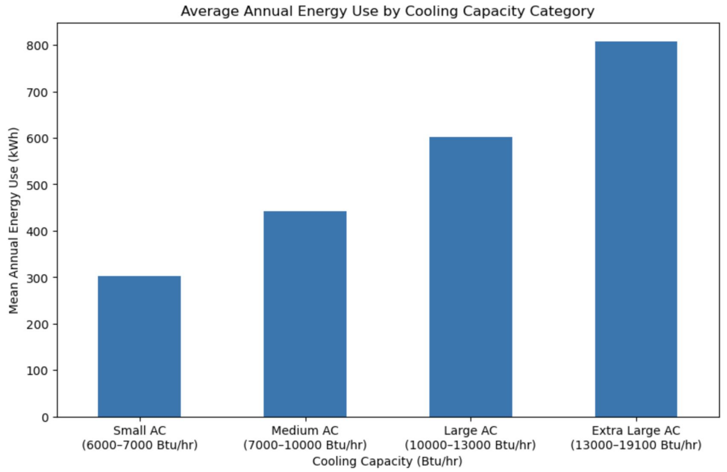




Air Conditioners Visualizations

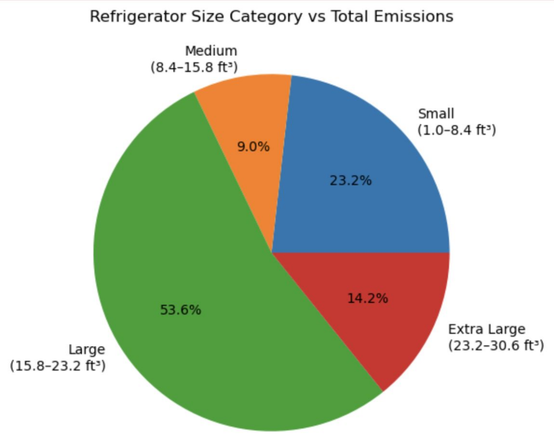
- **Data distribution assessed**: Pie charts revealed some imbalance with Bin 2 dominating emissions, but sufficient variation across capacity ranges for valid analysis
- **Hypothesis confirmed**: Bar chart demonstrated that average annual energy use increases with cooling capacity (BTU/hr)
- **Highest capacity = highest consumption**: Units with the greatest cooling capacity showed the highest mean kWh usage
- **Capacity drives energy demand**: Cooling power is a critical factor when evaluating air conditioner efficiency and environmental impact

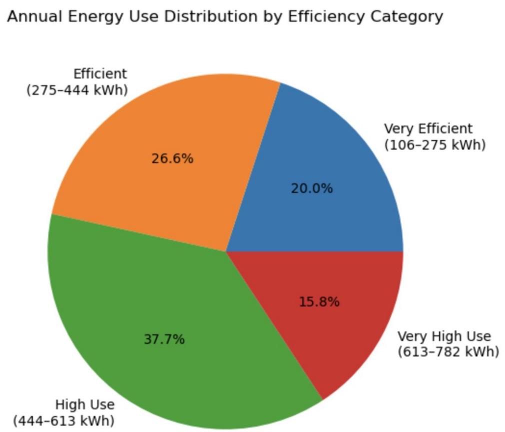


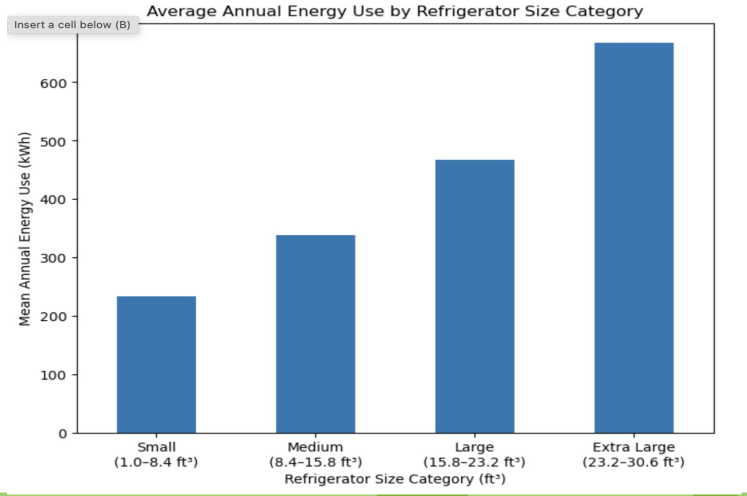


Refrigerators Visualizations

- **Data balance verified**: Pie charts showed fairly balanced distribution across size categories, with only slight underrepresentation in Bin 4
- **Clear upward trend**: Bar chart revealed that average annual energy use increases steadily with refrigerator size
- **Largest units consume most**: Bin 4 (largest refrigerators) demonstrated the highest average kWh per year
- **Size strongly predicts consumption**: Refrigerator size is a key factor for energy efficiency and consumer purchasing decisions







Ovens Visualizations

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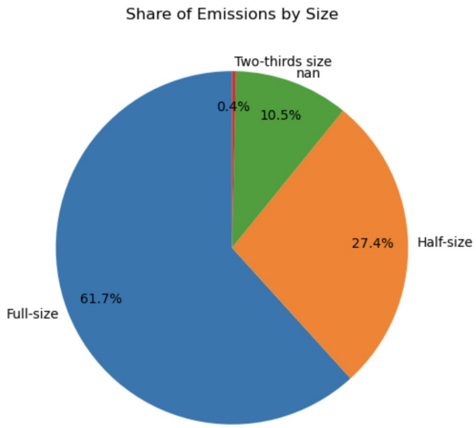
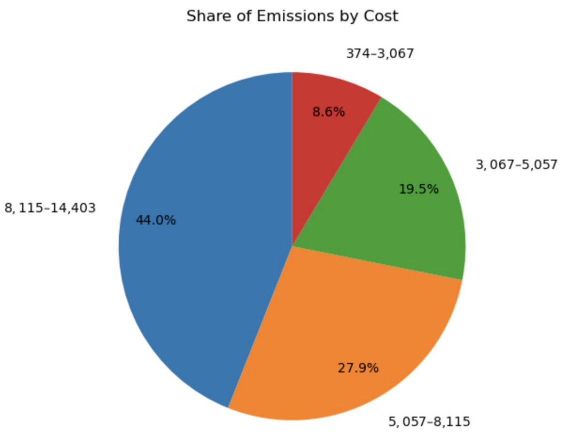
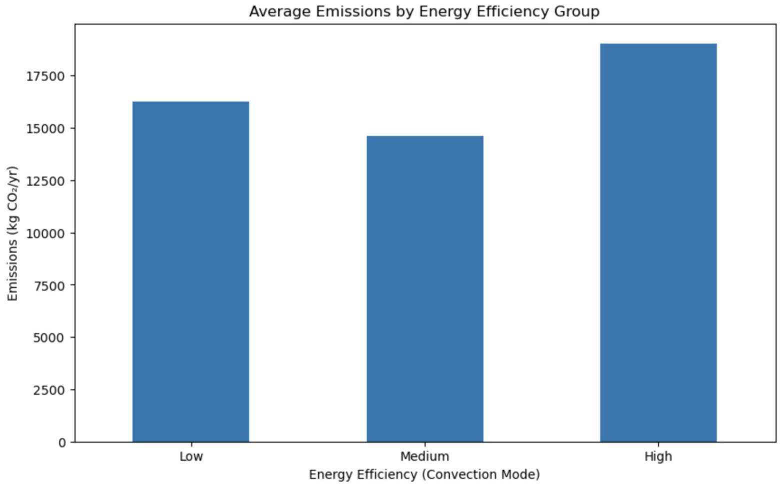
- **Data distribution checked**: Pie charts showed full-size ovens and middle-cost ranges dominate emissions, but sufficient variation exists across categories for analysis
- **Unexpected efficiency pattern**: High-efficiency ovens showed the highest average emissions, contradicting initial hypothesis
- **Medium efficiency performs best**: Medium-efficiency ovens demonstrated the lowest average emissions across all groups
- **Multiple factors at play**: Results suggest oven size and usage patterns may outweigh efficiency ratings in determining actual emissions







ML Analysis

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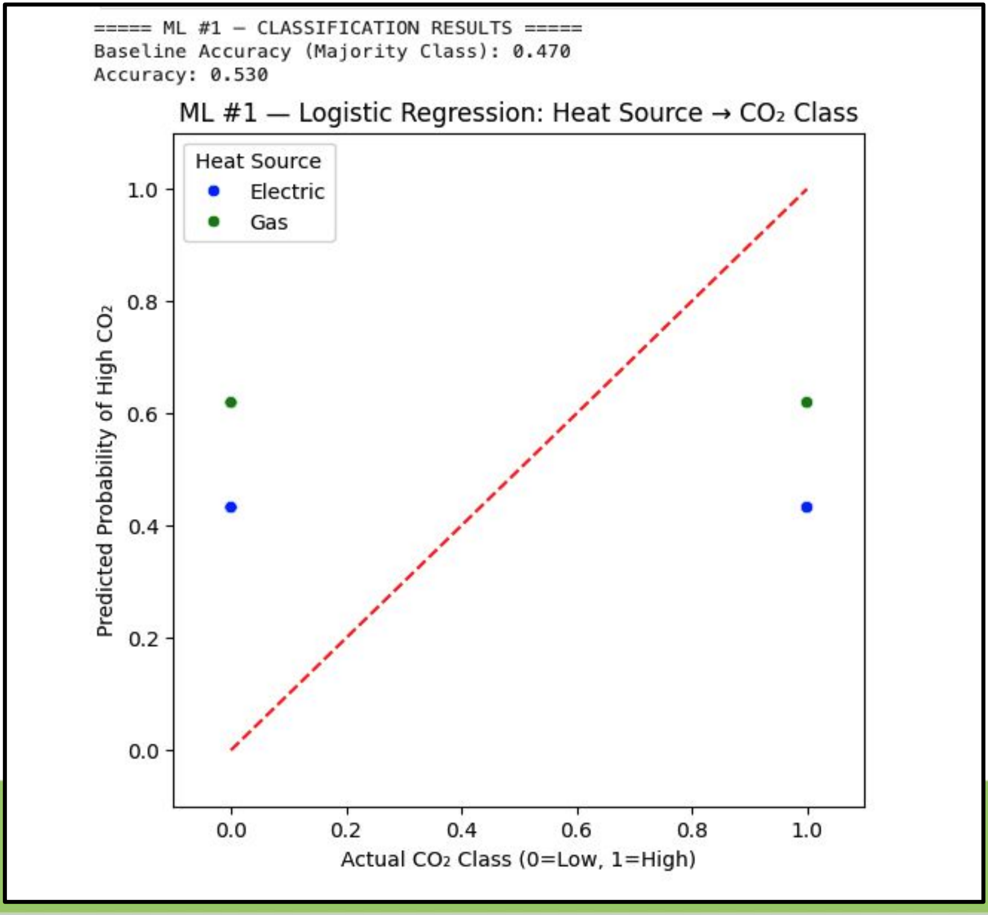
Attempt 1 with just heat source encoded `{"Gas": 1, "Electric": 0}`



ML Analysis

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Attempt #2: Using multiple features now

HeatSourceEncoded | Oven Classification | Size | Steam Generation Method

Oven Classification: Type I (Table, Countertop, Stand-mounted) & Type II (Floor, Roll-in)

Size: Full-size, Half-size, Two-thirds size

Steam Generation: Boiler-less / Spritzer / Injector, Boiler / Steam Generator

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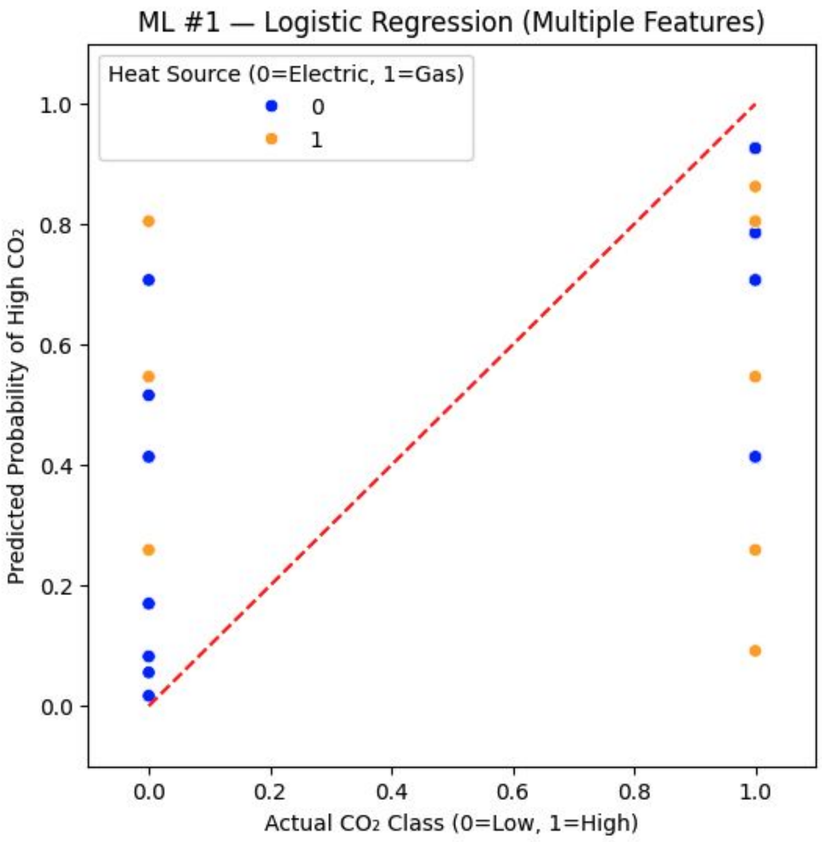
===== ML #1 — Improved Classification =====
Baseline Accuracy (Majority Class): 0.470
Model Accuracy: 0.687
```



Regression Results:

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===== ML #2 — REGRESSION RESULTS =====
Baseline MSE: 0.000092
Baseline R² Score: -0.126023
Model MSE: 0.000033
Model R² Score: 0.602264

Feature Coefficients:
  HeatSourceEncoded: 10.387342
  Oven Classification: -21.162393
  Size: 2.367043
  Baking Production Capacity (lbs/hr): 0.258500

===== Linear Regression Equation =====
InputRate_kW = 24.652274 + (10.387342)*HeatSourceEncoded + (-21.162393)*Oven Classification + (2.367043)*Size +
(0.258500)*Baking Production Capacity (lbs/hr)

===== CO₂ Emissions Equation (metric tons) =====
CO2 = 0.009713 + (0.004093)*HeatSourceEncoded + (-0.008338)*Oven Classification + (0.000933)*Size + (0.000102)*Bak
ing Production Capacity (lbs/hr)
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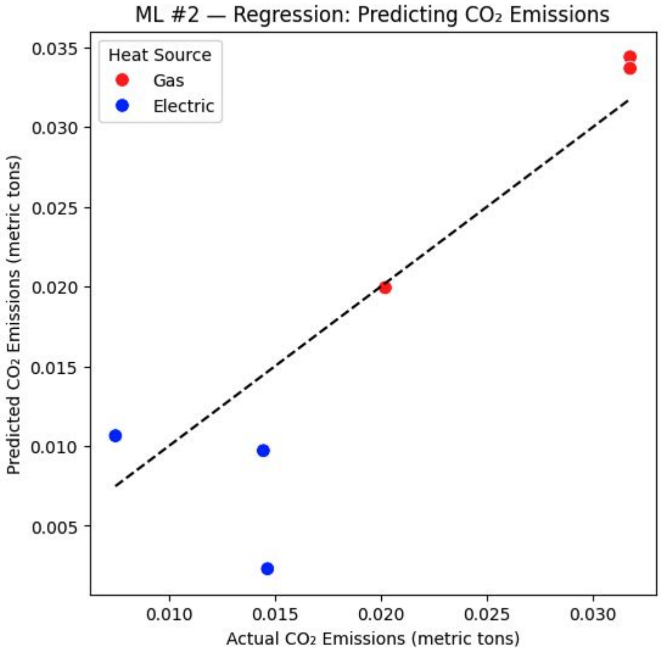

Sources

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1. https://www.energystar.gov/productfinder/advanced
2. https://www.epa.gov/energy/greenhouse-gas-equivalencies-calculator-calculations-and-references
3. https://blog.bismart.com/en/medallion-architecture

Thank You